■ Docker Learning Notes

Mental Model

```
Dockerfile
    ↓ build
Image (template)
    ↓ run
Container (running instance)

Image = blueprint/template
Container = running instance
One image can create many containers.
Think of a container as a tiny disposable Linux computer.
```

Dockerfile Instructions

```
FROM python:3.13-slim
WORKDIR /containerisedApp
COPY requirements.txt ./
RUN pip install --no-cache-dir -r requirements.txt
CMD ["python","turbineProcessing.py"]
```

RUN vs CMD

```
RUN: executes while building image.
CMD: executes when container starts.
```

Useful Commands

```
docker build -t turbine-app .
docker run turbine-app
docker run --rm turbine-app
docker run --name turbine-container turbine-app
docker run -it python:3.13-slim bash
docker images
docker ps
docker ps -a
docker rm CONTAINER_NAME
docker rmi IMAGE_NAME
docker system prune
```

Important Concepts

```
Images are immutable: edit code -> rebuild image -> run new container.
Don't edit running containers.
```

Volumes

```
docker run -v HOSTDIR:/containerisedApp image
Shares files between host and container.
```

Environment Variables

```
docker run -e API_KEY=value image
Python: os.getenv('API_KEY')
```

Port Mapping

```
docker run -p 5000:5000 image
localhost:5000 -> host 5000 -> container 5000
```

Docker Compose

```
Use docker compose up to start multi-container apps together.
```